

The Primordial Spectrum is Structural

scale-invariance from the dilation fixed point, the tilt as misalignment (ledger D8)

Evaluation note — FRC 21-gravity

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Abstract

The body reads the low large-angle microwave correlation as the signature of the wrapped chart but leaves the primordial spectrum “a direction, not a result.” This note makes it a result. Time is scale-dilation; the primordial fluctuations are the drive’s own jitter, so their spectrum is the *invariant* of scale-dilation — the unique scale-free Harrison–Zel’dovich spectrum, $n_s = 1$, at the symmetric quarter-turn. The longstanding near-scale-invariance is therefore the drive’s symmetry, derived without an inflaton or a transport mechanism. The observed red tilt $n_s - 1 \approx -0.035$ is the misalignment of the chart from that symmetric point, its exact magnitude the same chart-angle deviation that sets the cross-scale running of a_0 (Ω -hard, shared with D9). The wrapped chart truncates the two-point correlation beyond the horizon angular scale ($\sim 60^\circ$), predicting the observed low large-angle power and small $S_{1/2}$. The fixed point is exact; the data are tagged [data].

1 Scale-invariance is the fixed point of the drive

The dimensionless curvature power is $\Delta^2(k) = k^3 P(k)/2\pi^2$. For $P(k) \propto k^n$, $\Delta^2 \propto k^{n+3}$, so a scale-dilation $k \mapsto \lambda k$ leaves Δ^2 invariant iff $n = -3$, i.e. $\Delta^2 = \text{const}$, the Harrison–Zel’dovich spectrum $n_s = n + 4 = 1$.

Proposition 1 (Structural scale-invariance). *Time is scale-dilation: evolution is multiplication by the frame generator, $x \mapsto gx$. The primordial fluctuations are the fluctuations of that drive, so their spectrum is invariant under scale-dilation, which by the previous paragraph forces the scale-free form $n_s = 1$. The near-scale-invariance of the primordial spectrum — a coincidence in inflation, requiring a flat potential — is here the symmetry of the drive itself. No inflaton, no slow roll, no transport.*

2 The tilt is the chart misalignment (Ω -hard)

The exact $n_s = 1$ holds at the *symmetric* quarter-turn, the Fourier self-dual cardinal point of the meridian cycle. The actual observer chart sits slightly off it; the deviation is the spectral tilt. Its sign is red — the finite ultraviolet cutoff (a smallest substrate scale) removes short-wavelength power relative to long — and its exact magnitude is the chart-angle deviation, which is the same object as the cross-scale running of the acceleration scale $a_0(z)$ (the body’s Ω -hard residue). Thus

$$n_s - 1 \quad \text{and} \quad \frac{a_0(z)}{H(z)}$$

are two readings of one misalignment, and the exact tilt shares the Ω -hard status of that running.

3 The large-angle anomaly is the wrapped chart

Remark 1 (Truncation at the horizon). The chart is closed and wraps at the coherence horizon: super-horizon modes ($k < 1/R_H$) are *identified*, not populated. The two-point correlation therefore carries no independent power beyond the horizon angular scale, so $C(\theta) \simeq 0$ for $\theta > \theta_H \sim 60^\circ$. This is the observed large-angle anomaly — the low quadrupole and the anomalously small $S_{1/2} = \int_{-1}^{1/2} C(\theta)^2 d\cos\theta$ ($p \sim$ few percent) — now a prediction with the cutoff fixed at the horizon, not a fit.

The spectrum is the scalar fluctuation of one drive; there is no separate tensor source, so the tensor-to-scalar ratio is structurally small — a prediction rather than a tuning.

Status. Ledger D8 moves from **O** to **T|B**: the primordial spectrum is structural, with $n_s = 1$ derived as the dilation fixed point, the large-angle truncation derived from the wrapped chart, and a small tensor ratio predicted; the exact tilt magnitude is Ω -hard, one object with the a_0 running (D9). Checks in `validation/validate_primordial.py`.